MIL-F-44067C
2 April 1987
SUPERSEDING
MIL-F-44067B
30 January 1986

MILITARY SPECIFICATION

FRUITS, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized fruits in flexible pouches for use as a component of the Meal, Ready-to-Eat, Individual and Meal, Flight Feeding, Individual.

1.2 Classification. The fruits shall be of the following types (see 6.1):

Type I	- Applesauce, natural color and flavor, sweetened, regular style.
Type II	- Pineapple, sweetened, tidbits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8915

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SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products

U.S. Standards for Grades of Canned Applesauce

U.S. Standards for Grades of Canned Pineapple

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 N. 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Applesauce. The applesauce shall meet the requirements for natural color, and flavor type, sweetened, regular (or combined) style, U.S. Grade A of the U.S. Standards for Grades of Canned Applesauce.

3.2.2 Pineapple. The pineapple shall meet the requirements for tidbit style, U.S. Grade A of the U.S. Standards for Grades of Canned Pineapple. The liquid media for the pineapple shall be sugar (sucrose) syrup or natural pineapple juice.

3.2.3 Sugar, white. Sugar shall be white, refined, cane or beet sugar (sucrose) or a combination thereof. The sugar may be in the form of granulated or liquid syrup and shall comply with MIL-STD-900.

3.2.4 Acids. Ascorbic, citric, and malic acids shall comply with the Food Chemicals Codex.

3.2.5 Water. Water used for formulation, ice making and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Applesauce preparation. The pH of the applesauce product shall be adjusted to 4.0 or less, when necessary, by the addition of food grade malic or citric acid. To retard oxidation, 0.25 percent by weight of ascorbic acid shall be blended with the applesauce.

3.3.2 Pineapple preparation. The liquid medium for the pineapple shall be adjusted with refined sugar (sucrose) or sugar (sucrose) syrup, if necessary, such that the finished product will meet the Brix and drained weight requirements in 3.6. The pH of the pineapple product shall be adjusted to 4.0 or less, when necessary, by the addition of food grade citric acid. To retard oxidation, 0.25 percent by weight of ascorbic acid shall be blended with the pineapple.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 4 of MIL-P-44073.

b. A hot-fill process is permitted for applesauce (type I) provided the product is filled by a single-stage operation at a product temperature of not less than 195°F and, immediately after sealing, the pouches are oriented such that the hot fruit comes in contact with all interior surfaces of the pouch for such time as necessary to effect commercial sterility.

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c. Pineapple (type II) may be filled into pouches using a single-stage filling operation.

d. Each filled and sealed pouch (except those hot-filled) shall be in the retort process within 1 hour after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting at 215°F or lower until the internal temperature of the product reaches 195°F. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 4 of MIL-P-44073.

a. Fruit that has been thermostabilized by a hot-fill method (see 3.4b) is exempt from the retort processing requirement.

b. After completion of the thermostabilization operation (retort or hot-fill process), the filled and sealed pouches shall be held for 5 minutes and the rapidly cooled to an internal product temperature of 100°F or less.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 4.5 ounces.

e. No individual pouch shall contain less than 4.0 ounces of product.

f. No individual pouch shall contain less than 3.0 ounces of drained fruit (not applicable to applesauce).

g. The average drained weight of the fruit shall be not less than 3.5 ounces (not applicable to applesauce).

h. The pineapple shall meet the requirements for U.S. Grade B or better of the applicable U.S. Standards for Grades. Applesauce shall meet the requirements for U.S. Grade A of the U.S. Standards for Grades of Canned Applesauce.

i. The "cut-out" brix measurements for the liquid packing media of the pineapple shall be 18° or more, but less than 22°.

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j. The pH of the product in any individual pouch shall be not greater than 4.00.

k. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 4.0 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 4.5 ounces.

4.5.5 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermostabilization operation. The pouches shall then be inspected for the defects indicated in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

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TABLE I. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Drained weight of fruit in a pouch is less than 3.0 ounces. 4/ 5/
102	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g., dirt, insect parts, hair, wood, paper, paint, glass, or metal), foreign odor or flavor (e.g., burnt, scorched, stale, sour, rancid, mushy, or moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall only be performed when deemed necessary by an Agricultural Marketing Service (AMS) agent).

3/ The U.S. Grade of a lot shall be determined by the procedures set forth in the Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products. Failure to meet the finished product grade requirement specified in 3.6 shall be cause for rejection of the lot.

4/ The drained weight shall be determined in accordance with the drained weight method described in the applicable U.S. Standards for Grades. The drained weight shall be reported to the nearest 0.1 ounce and the lot shall be rejected if the sample average drained weight is less than 3.5 ounces.

5/ Not applicable to applesauce.

4.5.6 Brix testing (not applicable to applesauce). The contents of each pouch in the sample shall be individually tested for density of the packing media (degrees Brix) in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists utilizing test methods prescribed by 21 CFR Part 145. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The sample size shall be the number of sample units indicated by inspection level S-2. Any result failing to conform to the brix requirement specified in 3.6 shall be cause for rejection of the lot.

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4.5.7 pH testing. Three pouches shall be selected at random from the lot and individually tested for pH in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. The results shall be reported to the nearest 0.01. Any individual pouch having a pH greater than 4.00 shall be cause for rejection of the lot.

4.5.8 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of product required (see 1.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Fruit
Meal, Ready-to-Eat
Ration
Thermostabilized

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8915-0821

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-44067C	Fruits, Thermostabilized, For Meal, Ready-to-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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